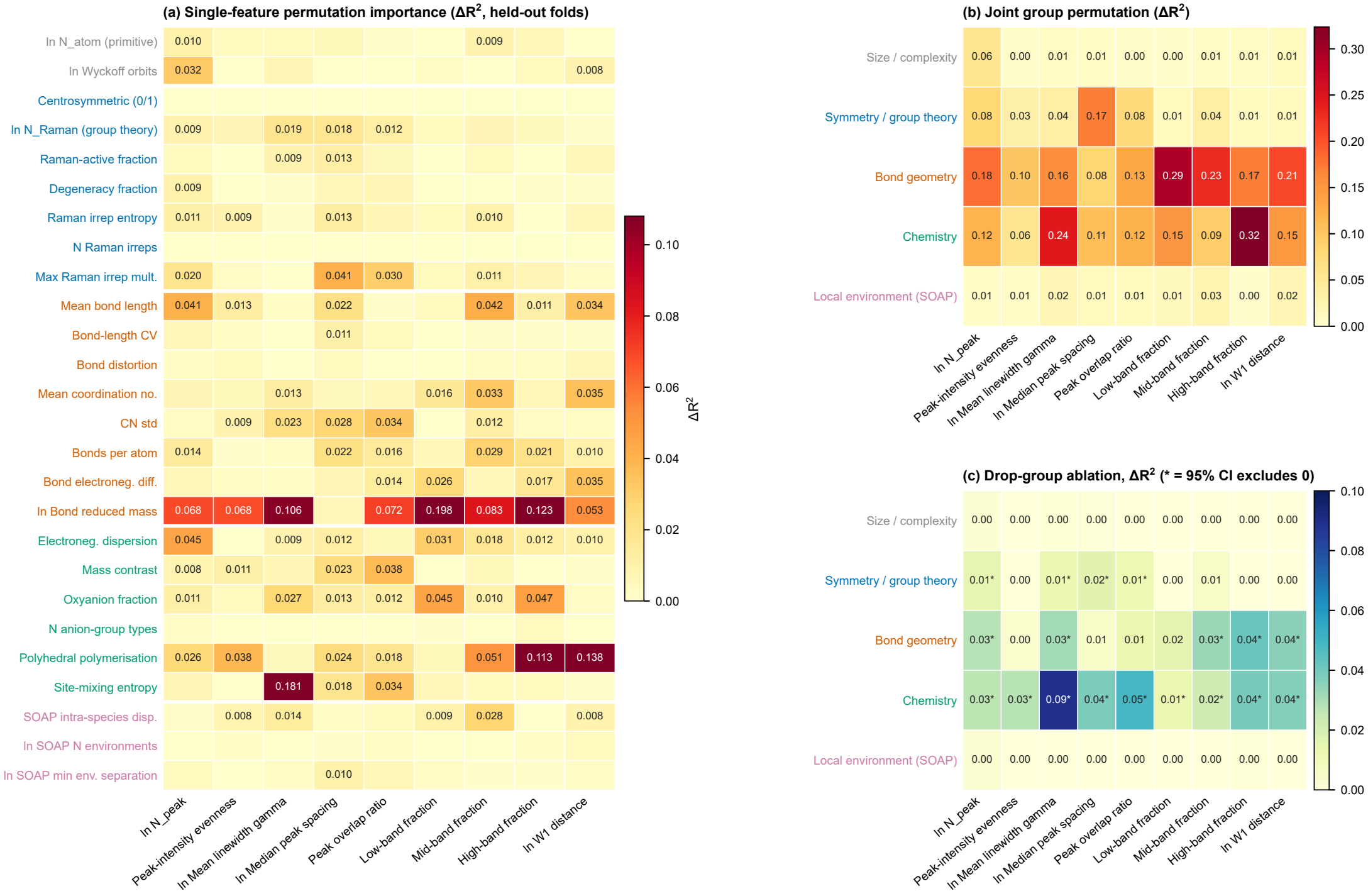

(a) Single-feature permutation (ΔR², held-out folds)
ln N_atom (primitive)
ln Wyckoff orbits
Centrosymmetric (0/1)
ln N_Raman (group theory)
Raman-active fraction
Degeneracy fraction
Raman irrep entropy
N Raman irreps
Max Raman irrep mult.
Mean bond length
Bond-length CV
Bond distortion
Mean coordination no.
CN std
Bonds per atom
Bond electroneg. diff.
ln Bond reduced mass
Electroneg. dispersion
Mass contrast
Oxyanion fraction
N anion-group types
Polyhedral polymerisation
Site-mixing entropy
SOAP intra-species disp.
ln SOAP N environments
ln SOAP min env. separation
ln N_peak
Peak-intensity evenness
ln Mean linewidth gamma
ln Median peak spacing
Peak overlap ratio
Low-band fraction
Mid-band fraction
High-band fraction
ln W1 distance
ΔR²
0.10
0.08
0.06
0.04
0.02
0.00
(b) Joint group permutation (ΔR²)
Size / complexity
Symmetry / group theory
Bond geometry
Chemistry
Local environment (SOAP)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
(c) Drop-group ablation, ΔR² (* = 95% CI excludes 0)
0.10
0.08
0.06
0.04
0.02
0.00